Software Testing, Deployment, and Maintenance Strategies

Module 1
Testing Strategies and Practices

Module 2
Software Deployment, Release, and Maintenance Strategies

- Deployment versus Release: Key Concepts and Responsibilities — Video • 5 min
- Enabling CI/CD Planning — Video • 4 min
- Release Checklist Template with Roles and Approvals — Video • 5 min
- Rollback Strategy Essentials — Video • 4 min
- Lab: Release plan with CI/CD workflows — Ungraded Plugin • 30 min
- Practice Quiz: Deployment and Release Management — Practice Assignment • Grade: 100%
- Activity: Role Play Scenario: Support the Release of Software Feature — Role Play • 15 min
- Data Migration Basics: Legacy to New Systems — Video • 5 min
- Data Mapping and Validation Techniques — Video • 5 min
- Data Mapping: Case Study — Video • 5 min
- Lab: Create a Data Migration Readiness Checklist — Ungraded Plugin • 30 min
- Practice Quiz: Data Migration and Validation — Practice Assignment • Grade: 100%
- Activity: Role Play Scenario: Data Migration Mapping and Validation — Role Play • 15 min
- Monitoring and Alerting; What to Track and Why — Video • 5 min
- Creating a Maintenance Calendar — Video • 6 min
- Reading: Retrospectives and Post-Mortem Templates — Ungraded Plugin • 4 min
- Lab: Design a Monitoring Plan and Maintenance Checklist — Ungraded Plugin • 30 min
- Activity: Role Play Scenario: Post Deployment Incident Response — Role Play • 15 min
- Practice Quiz: Post-Deployment and Continuous Improvement — Practice Assignment • Grade: 100%
- Activity: Podcast: Strategies for Software Deployment, Release and Maintenance — Ungraded Plugin • 15 min
- **Reading: Summary and Highlights: Deployment, Release, and Maintenance Strategies** — Reading • 3 min
- Glossary: Deployment, Release, and Maintenance Strategies — Ungraded Plugin • 5 min
- Graded Quiz: Deployment, Release, and Maintenance Strategies — Graded Assignment • 21 min

Module 3
Non-Functional Requirements

Module 4
Final Project, Final Exam, and Course Wrap-Up

Reading: Summary and Highlights: Deployment, Release, and Maintenance Strategies

Congratulations! You have completed this module.

At this point in the course, you know:

1. **Deployment and release** are two connected but distinct stages.
 - Deployment moves code into the production environment.
 - Release makes new functionality visible and accessible to users.
 - Separating them helps control timing, reduce risk, and allow safe testing.
 - Systems professionals manage checklists, confirm approvals, plan rollbacks, and coordinate timing between teams.
2. **Continuous Integration and Continuous Deployment (CI/CD)** automate how code is built, tested, and released.
 - CI merges code frequently into a shared repository with automated testing to detect issues early.
 - CD packages validated code for deployment or release.
 - Systems professionals define pre-deployment conditions, document decision points and rollback plans, manage timing, and lead stakeholder communication.
3. **Release checklists** ensure consistent, coordinated releases.
 - They define tasks, roles, and approvals across pre-release, release day, and post-release stages.
 - The Release Coordinator assigns responsibilities and verifies readiness.
 - Best practices include using a shared format, assigning owners, linking rollback plans, and reviewing with all teams.
4. **Rollbacks** restore systems to a stable state after failed or problematic deployments.
 - Types include application rollback (reverting code) and data rollback (restoring databases).
 - Systems professionals define rollback triggers, document the process, identify stakeholders, and conduct retrospectives to strengthen future releases.
5. **Data migration** moves information from old systems to new ones while preserving accuracy and meaning.
 - Responsibilities include mapping fields, running dry runs, validating results, and checking data after go-live.
 - Common mistakes include migrating outdated data, missing key fields, and skipping user validation.
 - Best practices involve backups, checklists, version control, and clear success criteria.
6. **Data mapping and validation** ensure that the transferred data is complete and correct.
 - Mapping templates capture source fields, target fields, and transformation rules.
 - Validation techniques include record count comparison, sample verification, format checks, and business rule validation.
 - Validation summaries compile results, include stakeholder sign-off, and confirm readiness for go-live.
7. **Monitoring and alerting** maintain system health after deployment.
 - Monitoring tracks uptime, response times, and error rates.
 - Alerts are classified by severity: critical, warning, or informational.
 - A monitoring and alerting matrix documents each metric, ownership, and trigger conditions, supporting faster incident response and long-term improvement.
8. **Maintenance calendars** structure ongoing system upkeep.
 - They list routine tasks, assign owners, and schedule timelines using tools such as spreadsheets or shared calendars.
 - Systems professionals identify tasks, determine frequency, and ensure accountability for each item.

Together, these practices form a complete system lifecycle framework that spans from deployment to long-term maintenance.

By mastering these areas: CI/CD, structured releases, rollbacks, data migration, validation, monitoring, and maintenance, systems professionals help organizations deliver reliable, secure, and high-performing systems that meet business needs and adapt safely to change.

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How can I help?

Explain this topic in simple terms
Give me a summary
Give me practice questions
Give me real-life examples

Ask me anything

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